Brokerage Operations Manager Take-Home

Prompt 1: four exceptions, an early-warning design and a working build. Prompt 2: the Q2 2026 back-office review.

Candidate	Rahul Das. Former FINRA analyst and Cboe cross-market investigator. SIE, CAPM.
Prototypes	rahul-das-456.github.io/Savvy (live demos and dashboard) · github.com/Rahul-Das-456/Savvy (code)
Data	Every account, name, order and number in the prototypes is fake.

Assumptions

- **Savvy is an introducing broker-dealer.** Fidelity (NFS) clears and carries on a fully disclosed basis under a FINRA Rule 4311 agreement. Fidelity runs the customer reserve formula; Savvy owns order entry, supervision, its own books and its FOCUS Part IIA.
- **Settlement is T+1** (SEA Rule 15c6-1). Times are Central. Fidelity's wire cut-off is earlier than Fedwire's 7:00 p.m. ET close.
- **Advisors are dually registered** or sit in Savvy's RIA, so block trades also follow the adviser's allocation policy.
- **AML alert decisions belong to the AML team.** As Ops Manager I gather facts, manage the advisor and escalate. I do not clear alerts myself.
- **"Litigation hold"** in Exception 4 is a legal order restricting trading. The late "FINRA/audit request" in Prompt 2 may be a Rule 8210 request; I treat it as one until Legal says otherwise.

Prompt 1 · Exception 1: Trade break

Account #12345 sold 500 AAPL at Fidelity. It held 450. Found after execution. Settles tomorrow (T+1).

Priority: fix today, before settlement

1. Immediate action

First: call Fidelity trade support to confirm the execution (500 shares, price, time) and the position (450 settled, nothing pledged or restricted). Then find out whether the other 50 shares exist anywhere.

- Pending activity on the account: unsettled buys, incoming ACATS, dividend reinvestment, a corporate action.
- The order ticket: who entered it, which channel, how it was marked (long), and whether any warning was overridden.
- The client's intent: did they mean 450 (full exit) or 500? Do they hold AAPL in another Savvy account?
- The cost: today's AAPL price against the 500-share execution price.

2. Resolution path

Option	When it works	Risk
A. Shares arrive by T+1	A pending buy or transfer settles by tomorrow. Document it and let the trade settle.	Only if confirmed today, not assumed.
B. Cancel/rebill + buy-in today	Rebill 450 to the client at the same price. Move the 50-share short to Savvy's error account and buy 50 AAPL today so 500 deliver on T+1.	Firm absorbs the price difference.
C. Journal from the client's other account	The client intended 500, owns more elsewhere, and authorizes the journal in writing.	Needs written authorization; same owner only.
D. Let it fail	Never. Fidelity buys in under Reg SHO 204 at the T+2 open at any price and charges Savvy.	Fail on the books plus a supervision finding.

Recommendation: A if Fidelity confirms the shares today, otherwise B. The error is ours, so the client should get exactly what they owned and lose nothing. B settles clean, keeps the cost in the firm, and documents the correction. The 50 shares were in effect a short sale marked long with no locate (**Reg SHO 200(g)(1), 203(b)(1)**), so the fix must close today. The cancel/rebill needs principal approval and a full order-memo trail (**FINRA 4515; SEA 17a-3(a)(6)**), and the CAT record gets corrected.

3. Stakeholder communication

> "Your AAPL sell for #12345 executed at 500 shares, but the account holds 450. We'll settle 450 for your client at the execution price, and the extra 50 is corrected in our error account at no cost to your client. Quick question so we close it out right: did you mean to sell all 450, or does the client hold AAPL somewhere else?"

Escalate to Compliance: yes, same day. It's a mismarked order under Reg SHO and an error-account trade, both of which Compliance logs. If the advisor has done this before (our fake data shows two in 30 days), it becomes a supervisory review under **FINRA 3110**.

4. Process improvement

- **Hard pre-trade check:** a sell cannot exceed settled shares, minus pledged and outgoing, plus confirmed buys settling by T+1. Overrides need a principal and are reported daily.
- Shares on an incoming ACATS count only once Fidelity confirms them.
- An error-account policy: principal sign-off on every trade, monthly P&L review by cause and advisor.
- A weekly break report by advisor and channel, with coaching after the second break in 30 days.

Exception 2: Duplicate cashiering entry

One $40,000 check was scanned twice by remote deposit, 15 minutes apart. Both processed. The account shows $80,000. The client hasn't noticed.

Priority: today, before any withdrawal

1. Immediate action

First: restrict withdrawal of the duplicate $40,000 at Fidelity. Then confirm it is one physical check.

- Both deposit images and IDs: routing, account and check number, amount, date, and who scanned each.
- Bank status: did the paying bank accept both presentments, or has it already rejected the second?
- Account activity since the credit: any trades or transfers funded by the extra $40,000.
- Whether Fidelity cashiering has an automatic reversal pending, so we don't reverse twice.

2. Resolution path

- **Reverse the duplicate by journal now**, maker and checker, both image IDs on the entry.
- Or wait for the bank to return the second item. That is slower, may bring return fees, and leaves the books and statement wrong in the meantime.
- If the extra cash was already used to buy securities, the client is in a cash shortfall: we tell the advisor and agree how to cover it.

Recommendation: reverse today, after confirming with Fidelity that no return is in flight. The blotter and customer ledger must show real receipts **(SEA 17a-3(a)(1), (a)(3))**, the phantom credit sits in Fidelity's reserve computation **(SEA 15c3-3(e))**, and the statement must be right **(FINRA 2231)**. Under **Reg CC 12 CFR 229.34(f)**, the depositary bank carries the duplicate-presentment risk, so fixing it ourselves first is cleaner.

3. Stakeholder communication

> "Your client's $40,000 check posted twice because of a scanning duplicate on our side. We've reversed the extra $40,000; the real deposit is untouched. Nothing for your client to do. It will show as a correction on the next statement. Sorry for the noise."

Escalate to Compliance: same day, as a notice, not a crisis. It is a books-and-records correction and a control gap. Escalate to AML only if the facts point away from a scanning error: for example, the client also deposited the same check at another bank. Duplicate presentment is a known fraud pattern **(FINRA 3310)**, so it gets ruled out, not assumed away.

4. Process improvement

- Hash each item on MICR line plus amount; reject a repeat scan for 90 days.
- Hold RCD funds as uncollected until the item clears.
- Daily reconciliation of RCD deposits against bank credits; a second person releases large items.
- Search the full quarter for other duplicates. One means the control is missing for all of them.

Exception 3: Wire held for AML review

$65,000 outbound international wire to a jurisdiction new to this client. Flagged and on hold. The advisor is frustrated and wants it sent today. I haven't read the alert yet.

Priority: answer the advisor in 5 minutes, decide on facts

1. Immediate action

First: reply to the advisor, then read the alert. Pressure from the advisor plays no part in the decision. The alert belongs to the AML team; my job is to get them what they need quickly.

- Which scenario fired, and the OFAC screening result on beneficiary, bank and country.
- Client profile: account age, stated purpose, usual wire activity, source of the $65,000 (recent deposit, liquidation).
- Beneficiary: name, bank, country, relationship to the client, stated purpose.
- Account-takeover signs: email, phone, address or password changes in the last 30 days; new devices or IPs.
- Client age. If 65+ and something doesn't fit, FINRA 2165 applies.
- A call-back to the client on the phone number on file before the request, never one supplied in it.

2. Resolution path

Outcome	When
Release	Call-back confirms it, purpose and beneficiary documented, fits the profile, no list match. AML closes the alert with a written rationale.
Request documents	Purpose or source of funds unclear. Wire waits; client provides support.
Escalate to the AML officer	Red flags remain after the call-back. SAR decision within 30 days of detection (**31 CFR 1023.320**).
Block or reject	OFAC match. Report within 10 business days (**31 CFR 501.603/.604**).
Temporary hold, FINRA 2165	Client is a specified adult and exploitation is suspected. Contact the trusted contact (**FINRA 4512**).

Recommendation: run the review now so a clean wire still makes today's cut-off, and nothing goes out on the advisor's say-so. Also confirm full originator and beneficiary details travel with the wire (**31 CFR 1010.410(f)**).

3. Stakeholder communication

> "Your client's wire is in a standard review that some international wires go through. I'm on it now and will update you by 2:00 p.m. CT. It will move faster if your client can confirm the purpose and their relationship to the recipient. I can't share detail on the review itself."

I never say "AML," "suspicious" or why it was flagged. SAR confidentiality (**31 CFR 1023.320(e)**) bars any hint that a report is under consideration. **Escalate to Compliance: immediately** on a possible OFAC match or a 2165 concern; **same day** if red flags survive the call-back. If it clears, AML documents the rationale and the wire goes out today.

4. Process improvement

- An international wire form at intake: purpose, relationship, source of funds. Fewer holds, faster clears.
- An automatic advisor notice within 30 minutes of any hold: status and next update time, no reason given.
- Alert SLAs by tier, with escalation at day 3 so nothing reaches day 5.
- A standard call-back script and a list of account changes that trigger a call.

Exception 4: Restriction applied to the wrong account

A litigation hold meant for #78901 went on #78091 by keying error. #78901 traded freely for two days. #78091's advisor asks why the client can't trade.

Priority: within the hour. Two live harms.

1. Immediate action

First: put the hold on #78901, with trading and outgoing money movement both blocked. A second person checks account name and TIN against the legal order. Assets may still be leaving.

- The legal order: what it restricts, from when, and whether it covers transfers as well as trading.
- All activity on #78901 during the two days: trades, transfers, checks, journals.
- The audit trail of the error: who keyed it, when, who approved.
- Any orders #78091 tried to place and had rejected.

2. Resolution path

- **#78901:** restrict now. Executed trades are not unwound on our own; Legal decides whether proceeds are frozen, reversed or reported to the court.
- **#78091:** lift the hold, approved and logged. If the client missed a trade, compute the make-whole against the price at rejection (**FINRA 2010**).

Recommendation: fix #78901 first, because that one may breach a legal order. Then release #78091 within the same hour. Both changes go through maker-checker with records under **SEA 17a-3, FINRA 4511 and 4512**.

3. Stakeholder communication

> To #78091's advisor: "This was our error. A restriction meant for a different account was put on your client's account by mistake. It was lifted at [time]. If your client tried to trade in the last two days and couldn't, send me the details and we'll review making them whole."

I don't mention the other account or the legal matter (privacy, **Reg S-P**). For #78901, Legal decides what the advisor is told. **Escalate to Compliance and Legal: immediately.** Legal decides on notice to the court or opposing counsel and whether a **FINRA 4530(b)** report is due. If #78091 complains in writing, it is logged under **FINRA 4513** and counted in 4530(d) statistics.

4. Process improvement

- Restrictions applied by account name and TIN, never by number alone, under maker-checker.
- A register of open legal orders, reconciled daily against restricted accounts.
- A confirmation report the next morning listing every restriction added or removed.
- Restrictions block money movement as well as trading by default.

5. Prevention: an early-warning system for Exception 1

I picked the trade break because it is the most preventable: every input exists before the order goes out, and the fix costs nothing if it happens before execution.

What it watches

- Settled positions and pledged or restricted shares (Fidelity position file).
- Unsettled buys and their settlement dates.
- Incoming and outgoing ACATS transfers and their status.
- Corporate actions with ex-dates near settlement.
- Account restrictions: legal holds, freezes.
- The live order stream, including sells earlier the same day.
- Each advisor's break history.

What triggers an alert

- **Block:** sell exceeds deliverable shares at T+1; account restricted.
- **Hold:** covered only by an unconfirmed transfer; possible duplicate (same account, symbol, quantity within 10 minutes).
- **Warn:** relies on an unsettled buy; corporate action within 2 business days.
- **Escalate:** advisor with 2+ oversell breaks in 30 days goes to the supervising principal.

Where the AI sits, and where it doesn't

- **Rules decide.** Blocking or releasing a client order must be explainable to an examiner line by line (**FINRA 3110**). No language model makes that call.
- **AI explains and learns.** Claude writes the morning brief for the supervisor and the plain-English advisor messages. It also compares each night's actual breaks with the alerts to find misses, and proposes rule changes for a human to approve.
- **Autonomy:** the check runs pre-trade on every sell (milliseconds, no AI in the path). A scheduled agent runs nightly over the day's orders and breaks, writes the brief, and opens tickets for repeat patterns.
- **Measured by:** oversell breaks per month (target 0), blocks overridden (false-positive rate), and minutes from block to corrected order.

6. Build it: Oversell Guard

A working Python script, standard library only, with an optional Claude step. Code, fake data and output: github.com/Rahul-Das-456/Savvy/tree/main/early-warning

Rule	Trigger	Result	Rule basis
R1	Account restricted (legal hold, freeze)	Block	FINRA 3110(b), 4511
R2	Same account, symbol and quantity within 10 minutes	Hold	FINRA 3110; SEA 17a-3(a)(6)
R3	Covered only by an unconfirmed incoming ACATS transfer	Hold	Reg SHO 200(g), 204(a)
R4	Sell > settled − pledged + confirmed buys by T+1 − outgoing transfers − earlier sells today	Block	Reg SHO 200(g)(1), 203(b)(1), 204(a)
R5	Sell is 90%+ of available	Info	FINRA 2090
R6	Corporate action goes ex within 2 business days	Warn	SEA 17a-3(a)(6)
R7	Sale relies on an unsettled buy	Warn	Reg SHO 200(g)(1)
R8	Advisor had 2+ oversell breaks in 30 days	Escalate	FINRA 3110(a)

Run on 10 fake sell orders (output shortened)

```
$ python3 oversell_guard.py
Oversell Guard  run 2026-11-05  settles 2026-11-06
10 sells | 5 blocked | 1 held | 2 warned | 2 clean

BLOCK ORD-1001  09:31:04  #12345   500 AAPL   R4 Sell exceeds deliverable position
      Sells 500, can deliver 450 by 2026-11-06. 50 shares short; the excess is not in the account.
BLOCK ORD-1003  09:35:12  #30442   300 TSLA   R4 Sell exceeds deliverable position
      Sells 300, can deliver 200 by 2026-11-06. 100 shares short; the excess is pledged as collateral.
BLOCK ORD-1006  09:47:30  #12345   150 MSFT   R4 Sell exceeds deliverable position
      Sells 150, can deliver 0 by 2026-11-06. 150 shares short; the excess is committed to an outgoing transfer.
BLOCK ORD-1007  09:52:18  #78901   900 VTI    R1 Restricted account
BLOCK ORD-1009  10:03:41  #55120    75 GOOGL  R4 Sell exceeds deliverable position
      Sells 75, can deliver 5 by 2026-11-06. 70 shares short; the excess is already sold earlier today.
HOLD  ORD-1005  09:44:01  #41907   220 AMZN   R3 Depends on unconfirmed transfer
HOLD  ORD-1009  10:03:41  #55120    75 GOOGL  R2 Possible duplicate (32s after ORD-1008)
WARN  ORD-1001 / 1005 / 1006              R8 Repeat advisor: A. Patel, 2 breaks in 30 days
WARN  ORD-1004  #22018  110 NVDA          R6 4-for-1 split ex 2026-11-06; R7 relies on unsettled buy
```

ORD-1001 is Exception 1 itself, caught before it reaches Fidelity. Each block also writes the advisor message, for example: *"Order ORD-1001 was not sent: it sells 500 AAPL but the account can deliver 450 by settlement. Resubmit for 450 or call Ops."* Results land in `out/alerts.json` (for a ticketing system) and `out/alerts.md` (for people).

The AI step

`--ai` sends the rules engine's output to Claude (`claude-opus-5-5`, low effort) with one instruction it cannot override: never change a decision. It returns a three-sentence summary, today's action items with owner and deadline, and patterns worth a control change. Without the SDK or credentials, the script skips this step and still runs.

What I'd build next

- Swap the CSVs for Fidelity's position and activity files, and hook R1 to R4 into the order-entry screen.
- A nightly job that matches real breaks to alerts and reports misses. Then the same pattern for Exceptions 2 to 4: a deposit hash, a wire checklist, name-plus-TIN checks.

Prompt 2 · 1. Data analysis

Three concerns, ranked by regulatory exposure first and advisor pain second.

Concern 1: regulatory obligations are slipping

- **FOCUS filed 4 days late.** Part IIA is due 17 business days after quarter end (**SEA 17a-5(a)**), so Q2 was due July 24, 2026. FINRA sees the filing date. Worse, the cause was a reconciliation break: if our books weren't current, **SEA 17a-11(d)** required same-day notice.
- **2 escheatment accounts past the state deadline.** State interest and penalties, and a common trigger for a state audit.
- **1 of 3 regulator requests answered late.** If it was a **FINRA 8210** request, lateness alone is a violation.

Why it matters most: three separate misses with one pattern. Nobody owns a compliance calendar with internal deadlines ahead of the regulatory ones.

Concern 2: AML alerts and wire holds

- **6 of 47 alerts (13%) took more than 5 business days.** 41 cleared in a day, so the six are different somehow. Aging alerts eat the 30-day SAR clock and are a standard **FINRA 3310** exam finding.
- **12 of 85 international wires held (14%), 3 escalated.** No advisor was told before they called. That is where the advisor frustration comes from.

Concern 3: trade processing under T+1

- **Breaks at 0.8% vs. 0.5% industry:** about 100 breaks against 62 expected, 38 extra.
- **22 block allocation discrepancies averaging 1.8 days to fix.** Under T+1 most were fixed after settlement: fail risk, and a post-trade allocation invites a cherry-picking question. Every change needs **FINRA 4515** approval.

Not a top concern yet: money movement averages 6 hours against a same-day target, so on average it passes. The 11 complaints say the tail is slow. I'd report the 90th percentile, not the mean.

2. Root cause assessment

Likely causes first, then the data I would pull before I call any of them a root cause.

Area	Likely root cause	Data I'd pull before concluding
FOCUS late	Reconciliation done only at quarter end; no internal deadline; one person owns it.	Open reconciliation breaks by age at June 30; when each was found; the FOCUS prep timeline by day; whether 17a-11 was considered.
Escheatment	Dormancy tracked by hand; state rules differ; due-diligence letters not on a calendar.	Last-contact dates for all accounts; the states and deadlines of the 2 missed; letter send dates; 17Ad-17 lost-holder search records.
Late regulator request	No request log with owner and due date; no extension asked.	Request type (8210 or exam); received, due and sent dates; where the delay sat.
AML aging	Capacity, or alerts waiting on client documents with no escalation rule.	The 6 alerts by scenario, analyst and wait reason; alerts per analyst; time in each status; SAR filing dates against detection dates.
Trade breaks	No pre-trade position check; allocations sent after trade date.	Breaks by cause, advisor, channel and time of day; override logs; allocation file rejects from Fidelity.
Wire turnaround	Rework on incomplete requests; manual review of routine recurring transfers.	Time per stage (intake, review, approval, Fidelity); rejection reasons; share of requests that recur to the same payee.

3. Regulatory exposure: disclosure and remediation

Find it, fix it, tell the regulator before they find it. FINRA gives credit for that (Regulatory Notice 19-23).

Late FOCUS

- Confirm the filing is now complete and correct; amend if the reconciliation changed any figure.
- FINOP and Compliance decide whether the books were "not current." If yes, the 17a-11(d) notice should have gone out; file it now and say so.
- Tell our FINRA coordinator proactively: what happened, why, and the fix. FINRA already has the late timestamp.
- Fix: daily Fidelity-to-ledger reconciliation, internal deadline at business day 10, FINOP sign-off at day 7.

2 escheatment accounts

- Try to reach the owners now. Contact before filing can still reactivate the account.
- File and remit to the states late, and use each state's voluntary disclosure program where offered to reduce penalties.
- Check the other 7 accounts were handled correctly, and run 17Ad-17 searches.
- Fix: a dormancy calendar by state, advisor outreach at 30 months, automatic due-diligence letters.

What I tell Legal and Compliance

A short written memo the same week, sent through Legal so the analysis stays privileged:

1. **Facts:** what was missed, the deadline, the actual date, the amount or accounts involved.
2. **Rules:** SEA 17a-5 and possibly 17a-11(d); the state unclaimed property acts and SEA 17Ad-17.
3. **Cause:** the root cause and whether it could affect anything else, such as other filings or other states.
4. **Exposure:** my view of the likely outcome, and the decisions I need from them: self-report or not, what goes to the FINRA coordinator, voluntary disclosure or not.
5. **Remediation:** the control fix, its owner and date, and how we will prove it works: two clean quarters.
6. **Records:** what we've preserved (recon workpapers, emails, filing receipts).

I state facts and options, and Legal and the CCO make the disclosure decision. I don't minimize, and I don't wait for the next quarter.

4. Advisor communication

A short note to all advisors, then the changes behind it. I'd send this from Ops leadership, not a ticket queue:

> "Thanks for telling us money movement feels slow and that holds catch you off guard. You were right on both. Starting [date]: (1) every request shows a live status, from received to in review to sent with the Fed reference; (2) if a wire is held for review, you'll get a message within 30 minutes with the next update time. We can't share why a review happens, but you'll never have to call to find out it exists. (3) Cut-offs: requests in by 1:00 p.m. CT go out same day; international wires need the purpose and recipient form. (4) Recurring transfers can move to a standing letter of authorization, which skips manual review. We'll report our 90th-percentile turnaround to you each month."

- **Transparent:** admit the gap, publish the cut-offs, and report the 90th percentile, slow tail included.
- **Realistic:** reviews will still happen and some wires will miss same-day. We promise notice and a time, not a release.
- **Within the rules:** hold notices never give a reason (**31 CFR 1023.320(e)**). Standing letters must meet the SEC's February 2017 custody-rule no-action conditions.

5. Monitoring: a daily agent instead of a quarterly review

A scheduled agent runs every morning at 7:00 a.m. CT. Deterministic checks find the problems. Claude writes the brief and drafts the escalations, and a named human owns every item.

Risk	What it watches	Flag (amber)	Escalate (red)
FOCUS reconciliation	Daily Fidelity vs. general ledger cash and positions; breaks by age; business days to the FOCUS deadline	Any break open 3+ days; under 10 business days to deadline with breaks open	Break open 5+ days (FINOP tests 17a-11(d) the same day); day 10 with the filing not ready goes to the CCO
Stalled AML alert	Alert age, last action date, status, SAR clock from detection date	No activity for 48 hours; alert at business day 3	Business day 5 goes to the AML officer; SAR clock at day 20 with no decision goes to the CCO
Escheatment deadline	Last client contact per account; each state's dormancy period and deadline; letter sent and returned dates	120 days before a deadline with no outreach; letter returned undeliverable	60 days out without a letter sent; 30 days out not filed goes to the CCO
Regulator requests	Request log: type, received, due, owner, extension	3 business days to due with no draft	1 day to due: ask for an extension or escalate
Trade breaks	Breaks per day, cause, advisor; allocations not done on trade date	Weekly rate over 0.5%; any allocation open at end of trade date	Any break open at the T+1 cut-off
Wires and money movement	Request status, time in stage, holds, notice sent	Request older than 8 hours; hold with no notice after 30 minutes	Same-day wire at risk 1 hour before cut-off

How it runs

1. **Collect:** nightly files from Fidelity, the general ledger, the AML case system, the CRM (last contact) and the filing calendar.
2. **Check:** the rules above, written as code, so the same inputs always give the same flags. That matters for exams.
3. **Brief:** Claude reads the flags and writes a one-page morning brief: what's red, who owns it, and what's due today. It drafts escalation emails but never sends to a regulator, client or advisor without a human.
4. **Route:** each flag opens a ticket with an owner and an SLA clock. Red goes to the CCO the same day.
5. **Learn:** each month, compare findings and misses with what the agent flagged, then tune the thresholds with Compliance's approval.
6. **Keep:** every flag, decision and rationale is retained under SEA 17a-4.

The live dashboard prototype shows this loop on fake Q2 data: risk indicators by month, an open-items queue with owners and SLA clocks, and a rule that you can't escalate or close without a written rationale. See rahul-das-456.github.io/Savvy/dashboard.html.

How I built this with Claude

Savvy asked us to use AI where it helps and to keep our own judgment in front. This is exactly how I split the work.

The tool

Claude Code (Claude Opus 5.5) in my terminal. It can read files, run code, search the web and push to GitHub. I also used its add-on skills for front-end design checks, PDF output and the Claude API.

The workflow

Step	What I did	What Claude did
Frame the problem	Gave it the brief, both prompts and the data. Set the assumptions (introducing firm on Fidelity, T+1) and the standard: exact SEC and FINRA citations, short answers.	Read the brief and flagged the parts I hadn't supplied yet: the PDF had only the cover pages, so it asked me for the scenarios and data.
Draft the answers	Set the context from my FINRA and Cboe surveillance work: how alerts age, what examiners ask for, where Reg SHO and 8210 bite. I own the final priorities and the advisor wording.	Drafted each answer with rule citations, in the structure I asked for.
Fake data	Required that every row be fake and that totals match the brief exactly.	Wrote a seeded Python generator so the totals reconcile: 12,450 trades, 100 breaks, 3,100 requests, 47 alerts, 85 wires.
Prototypes	Asked for a demo per exception and a monitoring dashboard. Reviewed them and redirected: its own repo, off my portfolio, named Savvy.	Built the pages, ran an automated design check at desktop, tablet and phone widths, clicked through every demo, and published to GitHub Pages.
Early-warning build	Asked for a real, runnable build, not a description. I own the design line that rules decide and AI only explains, which follows from exam expectations under FINRA 3110.	Proposed Exception 1 as the most preventable target. Wrote the Python rules engine and the optional Claude call, then ran it on fake orders. After the first run it fixed a noisy rule: near-full-sale notes were firing on orders already blocked.
This document	Set the questions and the order.	Laid out the answers and produced the PDF.

How I kept it honest

- **Citations:** I treat every rule cite as a claim to verify against the primary text (eCFR for SEC and BSA rules, the FINRA Rulebook), not as an answer.
- **Judgment calls stay human:** what to escalate, when, and what to tell an advisor. These come from how I've seen investigations and exams run.
- **Fake data only:** no real client, account or firm data went into the AI or the prototypes.
- **AI inside the product follows the same rule:** it drafts and summarizes, and people and deterministic rules decide.

Prototypes: rahul-das-456.github.io/Savvy · Code: github.com/Rahul-Das-456/Savvy · Rules cited as of November 2026.